LS Science Year 8 Programme of Study

Module One: Health and Lifestyle	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Working Scientifically content: **Topics:** ● 1.0 Working Scientifically	● Make predictions using scientific knowledge and understanding. ● Present observations and data using appropriate methods. ● Interpret observations and draw conclusions. ● Evaluate data, showing awareness of potential sources of error.	**7TWSa.05 8TWSa.05** Present and interpret observations and measurements appropriately. **9TWSa.05** Present and interpret results and predict results between the datapoints collected. **7TWSa.02 8TWSa.02** Describe trends and patterns in results, including identifying any anomalous results.	Pages 8-19
Health and Lifestyle content: ● Compare the effects of a healthy and unhealthy lifestyle on the human body. ● The human digestive system – structure and function. The products of digestion and how to test for them. **Topics:** ● 1.1 Nutrients ● 1.2 Food tests ● 1.3 Unhealthy diet ● 1.4 Digestive system ● 1.5 Bacteria and enzymes in digestion ● 1.6 Drugs ● 1.7 Alcohol ● 1.8 Smoking	● The content of a healthy human diet and why each constituent is needed. ● The consequences of imbalances in the diet, including obesity, starvation and deficiency diseases. ● The tissues and organs of the human digestive system, including adaptations to function. ● The effects of recreational drugs (including substance misuse) on behaviour, health and life processes. ● The impact of smoking on the human gas exchange system.	**8Bp.01** Identify the constituents of a balanced diet for humans as including protein, carbohydrates, fats and oils, water, minerals (limited to calcium and iron) and vitamins (limited to A, C and D), and describe the functions of these nutrients. **Note.** Organs of the human digestive system are covered in the Cambridge Primary Science curriculum. **8Bp.03** Discuss how human growth, development and health can be affected by lifestyle, including diet and smoking. **8Bp.02** Understand that carbohydrates and fats can be used as a store of energy in animals, and animals consume food to obtain energy and nutrients.	Pages 24-41
ASSIGNMENT ONE			

Module Two: The Periodic Table & Separation Techniques	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Periodic table content: ● What materials are like and why they behave as they do. **Topics:** ● 2.1 Three elements ● 2.2 Physical properties of metals and non-metals ● 2.3 Chemical properties of metals and non-metals ● 2.4 Groups and periods ● 2.5 The elements of Group 1 ● 2.6 The elements of Group 7 ● 2.7 The elements of Group 0	● The varying physical and chemical properties of different elements. ● The principles underpinning the Mendeleev Periodic Table and how it is organised. ● The Periodic Table: periods and groups; metals and non-metals. ● How patterns in reactions can be predicted with reference to the Periodic Table.	**7Cp.01** Understand that all substances have chemical properties and physical Properties. **7Cm.02** Know that the Periodic Table presents the known elements in an order. **7Cm.03** Know metals and non-metals as the two main groupings of elements. **9Cp.01** Understand that the groups within the Periodic Table have trends in physical and chemical properties, using group 1 as an example. **8Cc.06** Understand that some substances are generally unreactive and can be described as inert.	Pages 88-103
Separation techniques content: ● All about atoms and molecules and how we write chemical symbols and formulae. **Topics:** ● 2.8 Pure substances ● 2.9 Mixtures ● 2.10 Solutions ● 2.11 Solubility ● 2.12 Filtration ● 2.13 Evaporation and distillation ● 2.14 Chromatography	● The concept of a pure substance. ● Simple techniques for separating mixtures: filtration, evaporation, distillation and chromatography.	**8Cm.04** Know that purity is a way to describe how much of a specific chemical is in a mixture. **8Cp.02** Describe how paper chromatography can be used to separate and identify substances in a sample. **8Cp.01** Understand that the concentration of a solution relates to how many particles of the solute are present in a volume of the solvent. **8Cc.05** Describe how the solubility of different salts varies with temperature. **9Cc.03** Describe how to prepare some common salts by the reactions of metals with acids, and metal carbonates with acids, and purify them, using filtration, evaporation and crystallisation. **Note.** Separation of mixtures (filtration, evaporation) are covered in the Cambridge Primary Science curriculum.	Pages 106-120

Practical 1 - combustion	• **Make predictions using scientific knowledge and understanding.** • **Select, plan and carry out the most appropriate types of scientific enquiries to test predictions, including identifying independent, dependent and control variables, where appropriate.** • **Make and record observations and measurements using a range of methods.** • **Apply mathematical concepts and calculate results.** • **Present observations and data using appropriate methods, including tables and graphs.** • **Interpret observations and data, including identifying patterns and using observations, measurements and data to draw conclusions.**	**7TWSp.03 8TWSp.03 9TWSp.03 Make predictions of likely outcomes for a scientific enquiry based on scientific knowledge and understanding.** **7TWSp.04 8TWSp.04 Plan a range of investigations of different types, while considering variables appropriately, and recognise that not all investigations can be fair tests.** **7TWSc.02 8TWSc.02 9TWSc.02 Decide what equipment is required to carry out an investigation or experiment and use it appropriately.** **7TWSc.07 8TWSc.07 Collect and record sufficient observations and/or measurements in an appropriate form.** **9TWSc.07 Collect, record and summarise sufficient observations and measurements, in an appropriate form.** **7TWSa.05 8TWSa.05 Present and interpret observations and measurements appropriately.** **9TWSa.05 Present and interpret results, and predict results between the data points collected.** **7TWSa.02 8TWSa.02 Describe trends and patterns in results, including identifying any anomalous results.** **9TWSa.02 Describe trends and patterns in results, identifying any anomalous results and suggesting why results are anomalous.** **7TWSa.03 8TWSa.03 Make conclusions by interpreting results and explain the limitations of the conclusions.** **9TWSa.03 Make conclusions by interpreting results, explain the limitations of the conclusions and describe how the conclusions can be further investigated.**	
ASSIGNMENT TWO (Including Practical One)			

Module Three: Electricity and Magnetism	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Content: • What is electricity and how circuits work. • Magnetic fields and magnetic attraction and repulsion. • Electromagnets and their uses. **Topics:** • 3.1 Charging up • 3.2 Circuits and current • 3.3 Potential difference • 3.4 Resistance • 3.5 Changing the subject • 3.6 Series and parallel • 3.7 Magnets and magnetic field • 3.8 Electromagnets • 3.9 Using electromagnets	• Static electricity – separation of positive or negative charges when objects are rubbed together: transfer of electrons, between charged objects. • Electric current, measured in amperes. • Series and parallel circuits. • Potential difference, measured in volts. • The idea of electric field, forces acting across the space between objects not in contact. • Magnetic poles, attraction and repulsion. • Magnetic fields. • The magnetic effect of a current – electromagnets.	**7Pe.01** Use a simple model to describe electricity as a flow of electrons around a circuit. **7Pe.03** Know how to measure the current in series circuits. **7Pe.04** Describe how adding components into a series circuit can affect the current (limited to addition of cells and lamps). **9Pe.02** Know how to measure current and voltage in series and parallel circuits, and describe the effect of adding cells and lamps. **9Pe.01** Describe how current divides in parallel circuits. **9Pe.04** Use diagrams and conventional symbols to represent, make and compare circuits that include cells, switches, resistors (fixed and variable), ammeters, voltmeters, lamps and buzzers. **9Pe.03** Calculate resistance (resistance = voltage / current) and describe how resistance affects current. **8Pe.01** Describe a magnetic field, and understand that it surrounds a magnet and exerts a force on other magnetic fields. **8Pe.02** Describe how to make an electromagnet and know that electromagnets have many applications. **8Pe.03** Investigate factors that change the strength of an electromagnet.	Pages 164-184
ASSIGNMENT THREE			

Module Four: Biological Processes & Ecosystems	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Content: • Why do organisms need energy to function effectively and how do they make that energy. • Interdependence of organisms in an ecosystem. • Organisms are affected by their environment as they in turn affect the environment. **Topics:** • 4.1 Photosynthesis • 4.2 Leaves • 4.3 Plant minerals • 4.4 Aerobic respiration • 4.5 Anaerobic respiration • 4.6 Food chains and webs • 4.7 Disruption to food chains and webs • 4.8 Ecosystems • **Practical 2 – Population (National and Cambridge Curriculum links same as Practical 1)**	• The reactants in, and products of, photosynthesis. • The dependence of almost all life on Earth on photosynthetic organisms. • The adaptations of leaves for photosynthesis. • Aerobic and anaerobic cellular respiration. • The interdependence of organisms in an ecosystem. • How organisms affect, and are affected by, their environment, including the accumulation of toxic materials.	**9Bp.07** Know and use the summary word equation for photosynthesis (carbon dioxide + water -> glucose + oxygen, in the presence of light and chlorophyll). **8Bp.04** Know that aerobic respiration occurs in the mitochondria of plant and animal cells, and gives a controlled release of energy. **8Bp.05** Know and use the summary word equation for aerobic respiration (glucose + oxygen -> carbon dioxide + water). **7Be.02** Construct and interpret food chains and webs which include microorganisms as decomposers. **8Be.01** Identify different ecosystems on the Earth, recognising the variety of habitats that exist within an ecosystem. **8Be.02** Describe the impact of the bioaccumulation of toxic substances on an ecosystem. **8Be.03** Describe how a new and/or invasive species can affect other organisms and an ecosystem. **9Be.01** Describe what could happen to the population of a species, including extinction, when there is an environmental change. **9Bp.05** Know that plants require minerals to maintain healthy growth and life processes (limited to magnesium to make chlorophyll and nitrates to make protein).	Pages 40-54 Pages 56-62
ASSIGNMENT FOUR (including Practical 2)			

Module Five: Metals and other materials	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Content: • Chemical reactions between metals and other chemicals including displacement reactions. • Different materials and where they come from. **Topics:** • 5.1 Metals and acids • 5.2 Metals and oxygen • 5.3 The reactivity series • 5.4 Metal displacement reactions • 5.5 Extracting metals • 5.6 Ceramics • 5.7 Polymers • 5.8 Composites	• Reactions of metals with acids produce a salt and hydrogen. • Displacement reactions. • Properties of ceramics, polymers and composites.	**9Cc.03** Describe how to prepare some common salts by the reactions of metals with acids… **9Cc.04** Describe the effects of concentration, surface area and temperature on the rate of reaction, and explain them using the particle model. **9Cc.02** Identify examples of displacement reactions and predict products (limited to reactions involving calcium, magnesium, zinc, iron, copper, gold and silver salts). **7Cp.06** Understand that alloys are mixtures that have different chemical and physical properties from the constituent substances. **7Cp.07** Use the particle model to explain the difference in hardness between pure metals and their alloys. **9Cc.04** Describe the effects of concentration, surface area and temperature on the rate of reaction, and explain them using the particle model.	Pages 122–140
ASSIGNMENT FIVE			

Module Six: Energy	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Content: • Energy can be transferred in several ways: as a fuel, as heat through conduction, convection and radiation and from sources like the sun and wind into electricity. • Equations to link work, energy and power **Topics:** • 6.1 Food and fuels • 6.2 Energy resources • 6.3 Energy adds up • 6.4 Energy and temperature • 6.5 Energy transfer: particles • 6.6 Energy transfer: radiation • 6.7 Energy transfer: forces • 6.8 Energy and power • **Practical 3 – Insulation (National and Cambridge Curriculum links same as Practical 1)**	• Comparing energy values of different foods (from labels) (kJ). • Conservation of energy. • Heating and thermal equilibrium. • Comparing power ratings of appliances in watts (W, kW). • Comparing amounts of energy transferred (J, kJ, kW hour). • Domestic fuel bills, fuel use and costs. • Simple machines give bigger force but at the expense of movement (and vice versa).	**9Pf.04** Know that thermal energy will always transfer from hotter regions or objects to colder ones, and this is known as heat dissipation. **9Pf.05** Describe thermal transfer by the processes of conduction, convection and radiation. **9Pf.02** Describe the difference between heat and temperature. **9Cc.04** Describe the effects of concentration, surface area and temperature on the rate of reaction, and explain them using the particle model.	Pages 186-204
ASSIGNMENT SIX (including Practical 3)			

Module Seven: Adaptation and Inheritance	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Content • Different individuals of the same species will have different features. Variation between organisms and adaptations. • Evolution, natural selection and how organisms try to avoid extinction. **Topics:** • 7.1 Competition • 7.2 Adapting to change • 7.3 Variation • 7.4 Continuous and discontinuous • 7.5 Inheritance • 7.6 Natural selection • 7.7 Extinction	• A simple model of chromosomes, genes and DNA. • That there are differences between species. • The variation between individuals within a species is continuous or discontinuous. • Measurement and graphical representation of variation. • The variation between species and between individuals of the same species means some organisms compete more successfully, which can drive natural selection. • Changes in the environment may leave organisms less well adapted to compete successfully and reproduce, which in turn may lead to extinction. • The importance of maintaining biodiversity and the use of gene banks to preserve hereditary material.	**9Bs.03** Know that chromosomes contain genes, made of DNA, and that genes contribute to the determination of an organism's characteristics. **9Bp.03** Describe variation within a species and relate this to genetic differences between individuals. **7Bp.04** Use and construct dichotomous keys to classify species and groups of related organisms. **9Bp.04** Describe the scientific theory of natural selection and how it relates to genetic changes over time. **9ESs.01** Describe the consequences of asteroid collision with the Earth, including climate change and mass extinctions	Pages 64-68 Pages 70-82
ASSIGNMENT SEVEN			

Module Eight: The Earth	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Content: • Structure of the Earth, and the rocks of its crust. • The carbon cycle and climate change. • Earth's finite resources and recycling. **Topics:** • 8.1 The Earth and its atmosphere • 8.2 Sedimentary rocks • 8.3 Igneous and metamorphic rocks • 8.4 The rock cycle • 8.5 The carbon cycle • 8.6 Global heating • 8.7 Climate change • 8.8 Recycling	• The composition and structure of the Earth. • The rock cycle and the formation of igneous, sedimentary and metamorphic rocks. • The carbon cycle. • The production of carbon dioxide by human activity and the impact on climate. • Earth as a source of limited resources and the efficacy of recycling.	**7ESp.01** Describe the model of plate tectonics, in which a solid outer layer (made up of the crust and uppermost mantle) moves because of flow lower in the mantle. **9ESp.02** Explain why the jigsaw appearance of continental coasts, location of volcanoes and earthquakes, fossil record and alignment of magnetic materials in the Earth's crust are all evidence for tectonic plates. **7ESp.02** Describe how earthquakes, volcanoes and fold mountains occur near the boundaries of tectonic plates. **9ESc.01** Describe the carbon cycle (limited to photosynthesis, respiration, feeding, decomposition and combustion). **7ESp.03** Know that clean, dry air contains 78% nitrogen, 21% oxygen and small amounts of carbon dioxide and other gases, and this composition can change because of pollution and natural emissions. **8ESc.02** Understand that the Earth's climate can change due to atmospheric change. **9ESc.02** Describe the historical and predicted future impacts of climate change, including sea level change, flooding, drought and extreme weather events. **8ESc.01** Understand that there is evidence that the Earth's climate exists in a cycle between warm periods and ice ages, and the cycle takes place over long time periods. **8ESc.03** Describe the difference between climate and weather.	Pages 142-160
ASSIGNMENT EIGHT			

Module Nine: Motion and Pressure	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Content: • Using and understanding graphs of speed, distance and time. • Pressure in different substances and particle theory. • Turning forces (moments). **Topics:** • 9.1 Speed • 9.2 Motion graphs • 9.3 Pressure in gases • 9.4 Pressure in liquids • 9.5 Pressure in solids • 9.6 Turning forces	• Speed and the quantitative relationship between average speed, distance and time (speed = distance ÷ time). • The representation of a journey on a distance–time graph. • Relative motion. • Atmospheric pressure decreases with increase of height as weight of air above decreases with height. • Moment pressure in liquids, increasing with depth; upthrust effects, floating and sinking as the turning effect of a force. • Pressure measured by ratio of force over area – acting normal to any surface.	**8Pf.01** Calculate speed (speed = distance / time). **8Pf.02** Interpret and draw simple distance /time graphs. **8Pf.06** Use particle theory to explain pressures in gases and liquids (qualitative only). **8Pf.05** Explain that pressure is caused by the action of a force, exerted by a substance, on an area (pressure = force / area). **8Pf.04** Identify and calculate turning forces (moment = force x distance).	Pages 206-220
ASSIGNMENT NINE			